

Misallocation: A Crystal-Clear Beginner's Guide

From marginal products to Hsieh–Klenow, and on to the frontier

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1 The idea in one paragraph

An economy has a fixed amount of labour and capital *today*. Aggregate output depends not only on how much technology each producer has, but on *which producers get the inputs*. If a bad firm is hoarding workers that a good firm would use better, total output is lower than it could be — with the *same* total inputs and the *same* technologies. That shortfall is **misallocation**. It is a statement about the *allocation*, not about technology: no invention is needed to fix it, only reallocation. The literature’s central discovery (Restuccia and Rogerson 2008; Hsieh and Klenow 2009, henceforth HK) is that in developing countries the measurable footprint of misallocation is enormous — large enough that India’s manufacturing TFP could plausibly rise by 40–60% (HK’s estimate, moving to US-level allocative efficiency) just by reallocating inputs across existing plants.

2 The efficiency benchmark: equalised marginal products

Let firm i produce $Q_i = F_i(\mathbf{x}_i, A_i)$, and let the economy maximise an aggregator $U(Q_1, \dots, Q_N)$ subject to input constraints $\sum_i x_{im} \leq X_m$ for each input m (Donaldson, Lecture 6–7). At the optimal allocation $\{\mathbf{x}_i^*\}$, the planner’s first-order conditions require that the **value of the marginal product** of each input be equalised across all its users:

$$\underbrace{\frac{\partial U}{\partial Q_i} \frac{\partial F_i}{\partial x_{im}}}_{\text{VMP of input } m \text{ at firm } i} = \lambda_m \quad \text{for every firm } i.$$

Why? If firm 1’s marginal worker produces goods valued at 10 and firm 2’s at 6, moving one worker from 2 to 1 raises output by 4 at zero resource cost. Only when VMPs are equal is no such free lunch left. **Misallocation is occurring whenever the actual allocation gives a lower value of U than the optimal one** — equivalently, whenever VMPs are *dispersed*.

Worked Example 1 Two firms share $L = 10$ workers, produce with $Q_i = A_i L_i^{1/2}$, same output price = 1, $A_1 = 4$, $A_2 = 2$. Efficient allocation equalises $\text{MPL}_i = \frac{1}{2} A_i L_i^{-1/2}$: this gives $L_1/L_2 = (A_1/A_2)^2 = 4$, i.e. $L_1^* = 8, L_2^* = 2$, output $Q^* = 4\sqrt{8} + 2\sqrt{2} = 14.14$. Suppose instead a distortion forces the equal split $L_1 = L_2 = 5$: output = $4\sqrt{5} + 2\sqrt{5} = 13.42$. Same workers, same technologies, 5.1% of output lost. That 5.1% is misallocation.

*This guide draws on Dave Donaldson’s MIT 14.76/760 lectures “Misallocation I–II” (Spring 2026; copies in this folder), Hsieh and Klenow (2009), Restuccia and Rogerson (2017), and Hsieh, Klenow, Kyei Manu and Rockall (2026). All errors are the compiler’s.

3 Wedges: how economists describe the distortion

Whatever the underlying friction is, at the firm’s chosen input level it looks *as if* the firm faces a different effective input price. Define the **wedge** of firm i on input m :

$$\mu_{im} \equiv \frac{\text{VMP}_{im}}{w_m},$$

the ratio of the input’s value marginal product at the firm to the market price of the input. An undistorted profit-maximiser sets $\text{VMP} = w$, so $\mu = 1$. A firm that faces a tax, cannot borrow, is scared to grow, or exploits market power stops hiring while the VMP is still high: $\mu > 1$ (it is “too small”). A subsidised or protected firm over-expands: $\mu < 1$. **Dispersion in wedges across firms is exactly dispersion in marginal products — the signature of misallocation.** Sources (Donaldson’s list): taxes, subsidies and regulations; corruption and expropriation risk; credit constraints and asymmetric information; incomplete contracts and missing markets; *market power in output or input markets (monopoly, monopsony)*; externalities; and managerial mistakes. Note the important item for this thesis: markups and markdowns generate wedges too, so part of measured “misallocation” *is* market power.

4 The Hsieh–Klenow (2009) framework, step by step

HK make three assumptions that turn the abstract idea into something measurable with plant data.

(i) **Demand.** Each industry s aggregates its M_s firms’ varieties with CES:

$$Y_s = \left(\sum_{i=1}^{M_s} Y_{si}^{\frac{\sigma-1}{\sigma}} \right)^{\frac{\sigma}{\sigma-1}}, \quad \text{so } P_{si} \propto Y_{si}^{-1/\sigma}.$$

Downward-sloping demand means the efficient point is *not* “give everything to the best firm” — varieties are imperfect substitutes, so all firms produce, but better firms should be bigger. HK set $\sigma = 3$.

(ii) **Technology.** Cobb–Douglas with constant returns: $Y_{si} = A_{si} K_{si}^{\alpha_s} L_{si}^{1-\alpha_s}$, with the capital share α_s common within an industry (taken from US data, on the argument that US allocations are relatively undistorted).

(iii) **Distortions.** Each plant faces an output wedge $\tau_{Y_{si}}$ (scales all inputs down) and a capital wedge $\tau_{K_{si}}$ (distorts the K/L mix). Profit maximisation then gives the two marginal revenue products:

$$\text{MRPL}_{si} = \frac{\sigma-1}{\sigma}(1-\alpha_s) \frac{P_{si}Y_{si}}{L_{si}} = \frac{w}{1-\tau_{Y_{si}}}, \quad \text{MRPK}_{si} = \frac{\sigma-1}{\sigma}\alpha_s \frac{P_{si}Y_{si}}{K_{si}} = \frac{R(1+\tau_{K_{si}})}{1-\tau_{Y_{si}}}.$$

4.1 TFPQ versus TFPR: the distinction everything hinges on

- **TFPQ** (“Q” for quantity) is *physical* productivity: $\text{TFPQ}_{si} = A_{si} = \frac{Y_{si}}{K_{si}^{\alpha_s} L_{si}^{1-\alpha_s}}$. Firms legitimately differ in TFPQ — that is technology heterogeneity, not misallocation.
- **TFPR** (“R” for revenue) is *revenue* productivity: $\text{TFPR}_{si} = \frac{P_{si}Y_{si}}{K_{si}^{\alpha_s} L_{si}^{1-\alpha_s}} = P_{si}A_{si}$.

Here is the punchline that makes the whole literature work. With CES demand, a more productive (high-TFPQ) firm optimally produces more, which *lowers its price* P_{si} exactly enough

that — **absent distortions** — **TFPR is identical across firms in an industry**:

$$\text{TFPR}_{si} \propto (\text{MRPK}_{si})^{\alpha_s} (\text{MRPL}_{si})^{1-\alpha_s} \propto \frac{(1 + \tau_{Ksi})^{\alpha_s}}{1 - \tau_{Ysi}}.$$

TFPR is a geometric average of the firm’s marginal revenue products, i.e. of its *wedges*. So:

Dispersion of TFPQ = healthy heterogeneity. Dispersion of TFPR = mis-allocation.
A high-TFPR firm is a firm whose marginal products are high — a firm the economy is starving of inputs.

The workhorse statistic is therefore the **within-industry variance of log TFPR**. HK report it rising from the US to China to India; in this thesis’s ASI panel (2012–2022, $\sigma = 3$, within NIC-4×year) it is ≈ 0.55 – 0.66 , falling over the decade.

4.2 From dispersion to aggregate TFP gains

Industry TFP under the actual allocation is

$$\text{TFP}_s = \left[\sum_i \left(A_{si} \frac{\overline{\text{TFPR}}_s}{\text{TFPR}_{si}} \right)^{\sigma-1} \right]^{\frac{1}{\sigma-1}}, \quad \text{TFP}_s^* = \left[\sum_i A_{si}^{\sigma-1} \right]^{\frac{1}{\sigma-1}},$$

where $\overline{\text{TFPR}}_s$ combines value-added–share–weighted harmonic means of MRPK and MRPL. The **gain from full reallocation** is $Y^*/Y = \prod_s (\text{TFP}_s^*/\text{TFP}_s)^{\theta_s}$. If TFPR were lognormal, $\ln(Y^*/Y) \approx \frac{\sigma}{2} \text{var}(\ln \text{TFPR})$ — gains grow with dispersion *and* with σ (closer substitutes make misallocation costlier). HK’s headline: reallocating to US-level efficiency would raise manufacturing TFP by roughly 40–60% in India and 30–50% in China.

4.3 The measurement recipe (what you actually do with plant data)

For each plant: value added $P_{si}Y_{si}$, wage bill wL_{si} , capital stock K_{si} . Then, per industry-year:

1. Set σ (e.g. 3), R (e.g. 0.10), α_s from US industry shares.
2. $\text{MRPL} \propto \text{VA}/L$; $\text{MRPK} \propto \text{VA}/K$; wedges from the FOCs; $\text{TFPR} \propto \text{VA}/(K^\alpha L^{1-\alpha})$; $\text{TFPQ} \propto \text{VA}^{\sigma/(\sigma-1)}/(K^\alpha L^{1-\alpha})$ (the $\sigma/(\sigma-1)$ exponent undoes the price).
3. Trim tails (HK trim 1% of both TFPQ and TFPR deviations) — results are sensitive to outliers.
4. Compute $\text{var}(\ln \text{TFPR})$ within industry; compute the gains formula.

5 The caveats a beginner must know (Donaldson, Lecture 8)

Measured TFPR dispersion is an *upper bound* on policy-relevant misallocation, for at least five reasons.

1. **Technology assumptions.** HK assume constant returns ($\gamma_i = 1$). If the scale elasticity $\gamma_i(L_i)$ is unknown, the wedge cannot be separated from returns to scale using revenue and cost data alone — an identification problem; relaxing CRTS (Carrillo et al. 2023 route) changes measured wedges.
2. **Measurement error** in plant revenue or inputs inflates dispersion mechanically (Bils, Klenow and Ruane 2021 propose a fix using panel responsiveness; they find a large share of measured US–India dispersion is error).

3. **Adjustment costs.** Plants hit by shocks cannot adjust instantly; Asker, Collard-Wexler and De Loecker (2014) show volatility plus time-to-build generates TFPR dispersion with no policy distortion at all.
4. **Markups and markdowns.** With variable markups, P_{si} does not fall one-for-one with A_{si} ; part of TFPR dispersion is market power, which has different welfare implications than taxes. (This thesis quantifies that component for India: small, $\sim 2\%$ of dispersion — Chapter 5, following Liang 2023.)
5. **Which reform?** The gains formula answers “remove *all* wedges,” which is not a feasible policy. Targeted reforms remove only components of dispersion — see the frontier below.

6 The frontier: misallocation versus inequality (HKKR 2026)

Hsieh, Klenow, Kyei Manu and Rockall (2026) ask why size-dependent misallocation persists in India, using ASI + informal-sector (SUM) data with a model where firms differ also in *skill-biased* productivity and workers are skilled/unskilled. Their findings: neutralising the *size* component of wedges raises output only 1–4%, but raises the skill premium 10–15 points (large firms are skill-intensive), cuts low-skill wages $\sim 7\%$, and lowers consumption-equivalent welfare by 8–11% — roughly 90% of agents lose. Size-dependent policies may persist because removing them is distributionally toxic. Two lessons for practice: (a) *decompose* dispersion — the size-systematic part of Indian organised-sector TFPR dispersion is only 2–6% (this thesis, 2012–2022), so most formal-sector misallocation is *within*-size-class; (b) removing *all* dispersion, in contrast, raises both skilled and unskilled wages in their model — broad factor-market reform and targeted size-reform are very different objects.

7 How this thesis implements the framework (Chapter 5 road map)

ASI factory panel 2012–2022 (423k plant-years) $\rightarrow$ HK cleaning at $\sigma = 3 \rightarrow$ within NIC-4 $\times$ year dispersion of TFPR/MRPL/MRPK (capital wedges dominate) $\rightarrow$ Kimball variable-markup demand calibrated at 2017 ($\hat{\sigma} = 5.34$, superelasticity 0.183) $\rightarrow$ counterfactual gains: market 175%, planner 179%, CES 189% $\rightarrow$ market-power share of dispersion $\sim 2\%$ $\rightarrow$ robustness: per-year, own- α , demand systems, tobacco, gross-output vs value-added (VA overstates misallocation 4–7 $\times$), size-dependent decomposition (2–6%). Every number lives in E:\ASI PRATP Panel\ObjB.Misallocation\ with a do/log pair.

8 Annotated reading list

- **Start here:** Restuccia, D. and R. Rogerson (2017), “The Causes of Misallocation,” *Journal of Economic Perspectives* 31(3), 151–174. The gentlest survey. [doi:10.1257/jep.31.3.151](https://doi.org/10.1257/jep.31.3.151)
- **The classic:** Hsieh, C.-T. and P. Klenow (2009), “Misallocation and Manufacturing TFP in China and India,” *QJE* 124(4), 1403–1448. Read Sections I–III slowly; everything above is there. [doi:10.1162/qjec.2009.124.4.1403](https://doi.org/10.1162/qjec.2009.124.4.1403)
- **The founding model:** Restuccia, D. and R. Rogerson (2008), “Policy Distortions and Aggregate Productivity with Heterogeneous Establishments,” *Review of Economic Dynamics* 11(4), 707–720.
- **Lectures (local copies in this folder):** D. Donaldson, MIT 14.76/760, “Misallocation I” (Lectures 6–7: definitions, wedges, HK09) and “Misallocation II” (Lecture 8: relaxing

CRTS, identification). Crystal-clear graduate treatment; this guide follows their arc.

- **The critiques:** Asker–Collard–Wexler–De Loecker (2014, *JPE*) on adjustment costs; Bils–Klenow–Ruane (2021, *JME*) on measurement error; Haltiwanger–Kulick–Syverson (2018, NBER WP 24683) on the model’s dispersion diagnostics.
- **Market power link:** Liang (2023, *Review of Economic Dynamics*) — variable markups inside HK; the template for this thesis’s Chapter 5. Edmond–Midrigan–Xu (2023, *JPE*) on how costly markups are.
- **The frontier:** Hsieh, Klenow, Kyei Manu and Rockall (2026), “Misallocation versus Inequality,” working paper (copy in this folder). Distributional consequences of removing misallocation.
- **Video:** search YouTube for recorded lectures by Chang-Tai Hsieh, Peter Klenow, or Diego Restuccia on “misallocation” (e.g. conference keynotes and central-bank lecture series); Donaldson’s MIT course materials are the closest match to this guide.

One-page cheat sheet

Object	Meaning
$\text{TFPQ}_{si} = A_{si}$	Physical productivity; heterogeneity is fine
$\text{TFPR}_{si} = P_{si}A_{si}$	Revenue productivity; equalised if no distortions
$\mu_{im} = \text{VMP}_{im}/w_m$	Wedge; $\mu > 1$ firm too small, $\mu < 1$ too big
$\text{MRPL} \propto \text{VA}/L$	High where labour is scarce (wedged)
$\text{MRPK} \propto \text{VA}/K$	High where capital is scarce; disperses most in India
$\text{var}(\ln \text{TFPR})$	The misallocation statistic (within industry)
$\ln(Y^*/Y) \approx \frac{\sigma}{2} \text{var}(\ln \text{TFPR})$	Lognormal gains approximation
Caveats	CRTS, measurement error, adjustment costs, markups, feasibility
Frontier	Who gains from removing it? (HKKR: often, a minority)